

Kirill Grinko

Education

- Sep 2023 – Present **Moscow Institute of Physics and Technology**, *finished 5th semester bachelor*,
Overall GPA: 4.75/5; Computer science courses GPA: 4.90/5
System Programming and Applied Mathematics, Phystech School of Applied Mathematics and Informatics.
- 2019 – 2023 **Moscow State School 57**, *grades 8 – 11*, GPA: 5/5
Focus on physics and math. Graduated with federal and Moscow gold medals.

Experience

- Sep 2025 – Dec 2025 **Software Engineering Intern at VK**, *database team*
Optimized RAM usage for a high-throughput, concurrent in-memory key-value store. Leveraged an external NVMe device as an additional storage layer and designed the algorithms governing interaction between the in-memory engine and the disk, implementing efficient data offloading, retrieval and synchronization to maximize throughput and minimize memory pressure.
- Spring 2026 **Compilers course at MIPT**, *GitHub repo*
Built a compiler for a small functional language using C++23. Implemented the compilation pipeline: lexing, parsing, AST construction, semantic analysis, type checking, LLVM IR generation, object emission, and executable linking. Supported language features including algebraic data types, pattern matching, function definitions, and function calls.
- Spring 2025 **Concurrency course at MIPT**
Implemented various synchronization primitives using atomic operations only. Built a thread pool and stackful coroutines. Combined them to create fibers (user-space cooperative threads) and implemented synchronization primitives for them. Developed functional combinators for working with futures. Implemented a lock-free data structures (atomic shared_ptr, stack, queue) using the hazard pointers scheme. Optimized thread pool with reducing count of dynamic allocations and implementation of work-stealing scheduling strategy.
- Fall 2024 – Spring 2025 **C++ course at MIPT**, *GitHub repo*
Implemented template allocator-aware data structures (unordered_map, list, smart pointers, strategy-based array, matrix), type-erased configuration system with vtable, compile-time 8-puzzle solver, JSON converter, geometry primitives, big_integer.
- Fall 2024 – Spring 2025 **Algorithms and data structures course at MIPT**, *GitHub repo*
Implemented solutions to competitive programming problems covering fundamental algorithms and data structures, dynamic programming techniques, graph algorithms, algorithms on strings, and number theory algorithms.

Skills

- Hard skills Algorithms & data structures, Concurrency, C++, C, Assembly x86 & ARM, Python, Haskell, LaTeX, Git, Bash, Docker.
- Soft skills Quick-learning, Hard-working, Outgoing and collaborative.
- Languages English (C1), Russian (native speaker).

Achievements

- 2023 **All-Russian Olympiad in physics**, *final stage participant, top 80 in country*
- 2023 **Phystech (MIPT) Olympiad in physics**, *final stage gold medal*
- 2022 **Rosatom Olympiad in physics and maths**, *final stage gold and silver medals*
- 2019 **International Experimental Physics Olympiad**, *bronze medal*
- 2019 **Maxwell Physics Olympiad**, *final stage silver medal*